

Differential effects of combined donepezil and language therapy on word and pseudoword reading in post-stroke aphasia

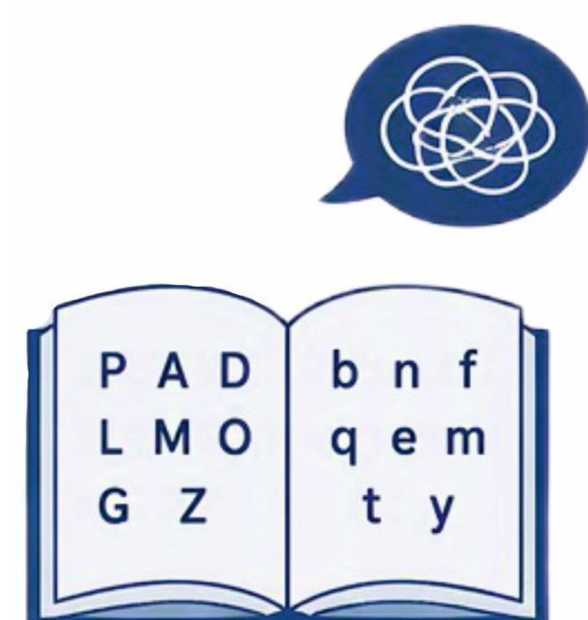
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BACKGROUND

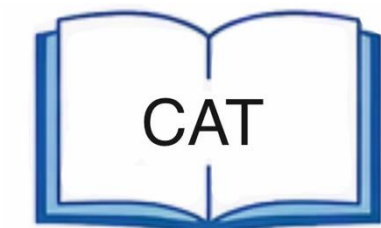
1. THE PROBLEM

Post-stroke aphasia (PSA) often causes persistent reading impairments that limit everyday communication and functioning.



2. READING RELIES ON TWO PARTLY DISTINCT MECHANISMS*

WORD READING (lexical route)



Relies on word-specific orthographic and phonological memory representations

PSEUDOWORD READING (sublexical route)



Relies on grapheme-phoneme conversion rules

These mechanisms may show different patterns of recovery.

*Coltheart et al. 2001

3. THE INTERVENTION AND THE GAP



DONEPEZIL, an acetylcholinesterase inhibitor, has shown **benefits** in chronic PSA both alone and in combination with **intensive language therapy** (Berthier et al., 2006; 2014).

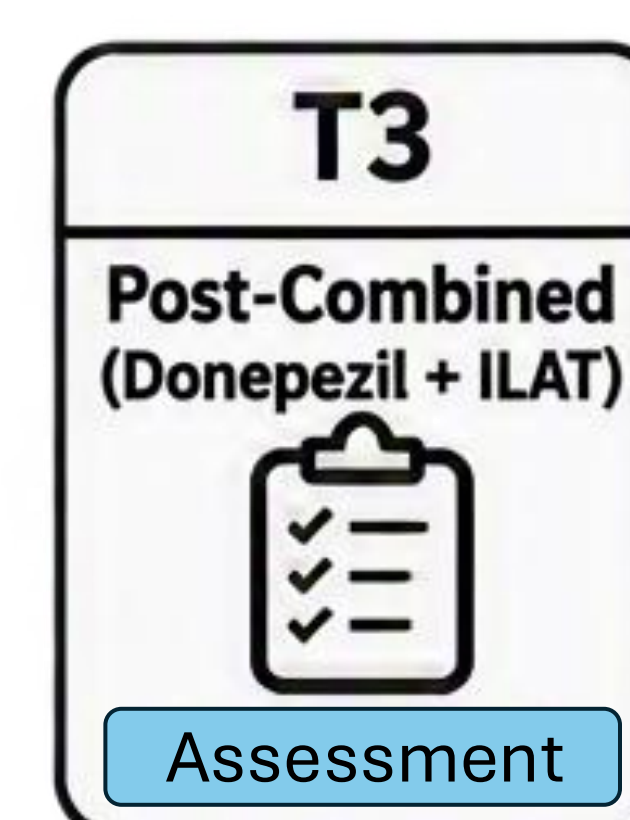
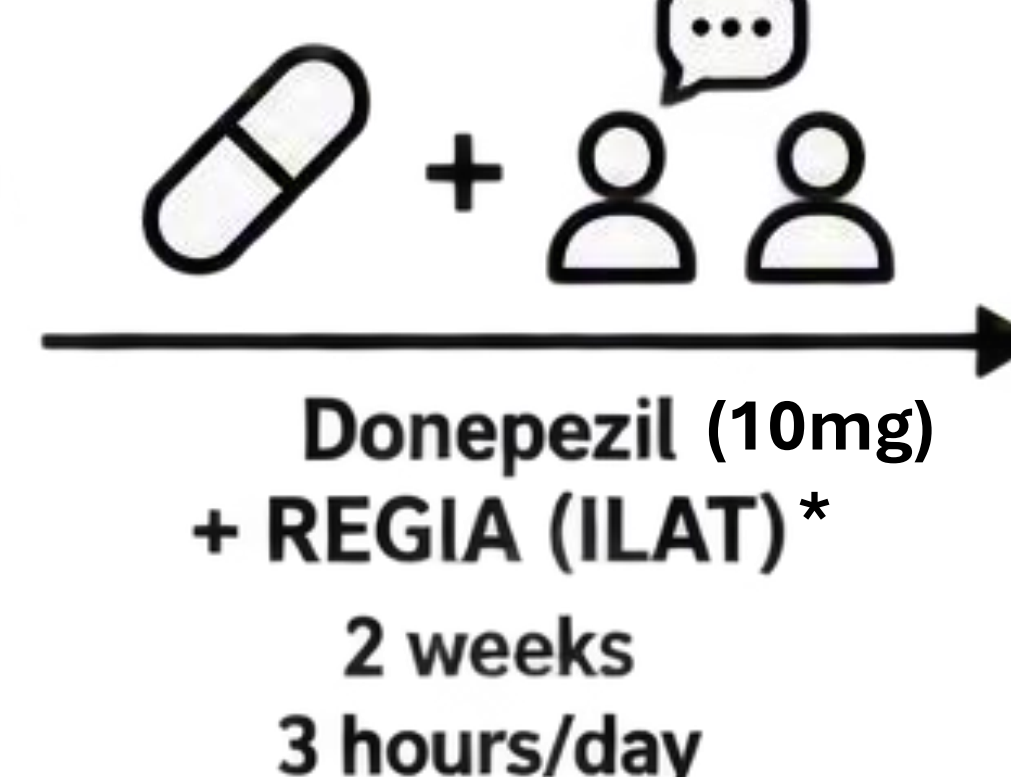
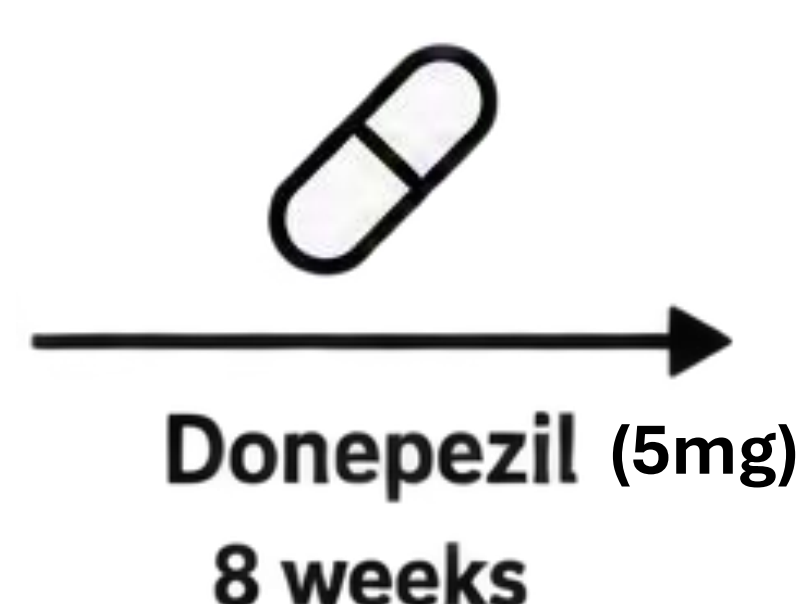
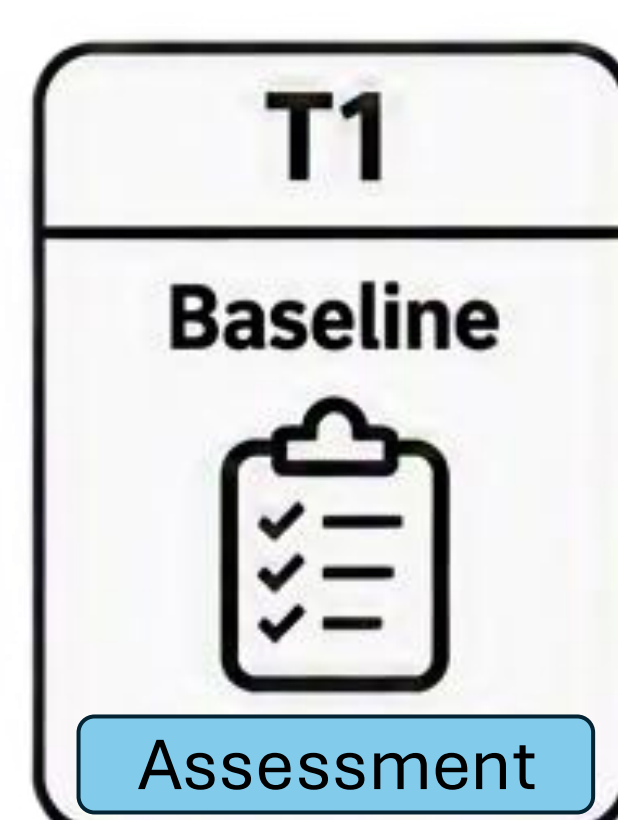
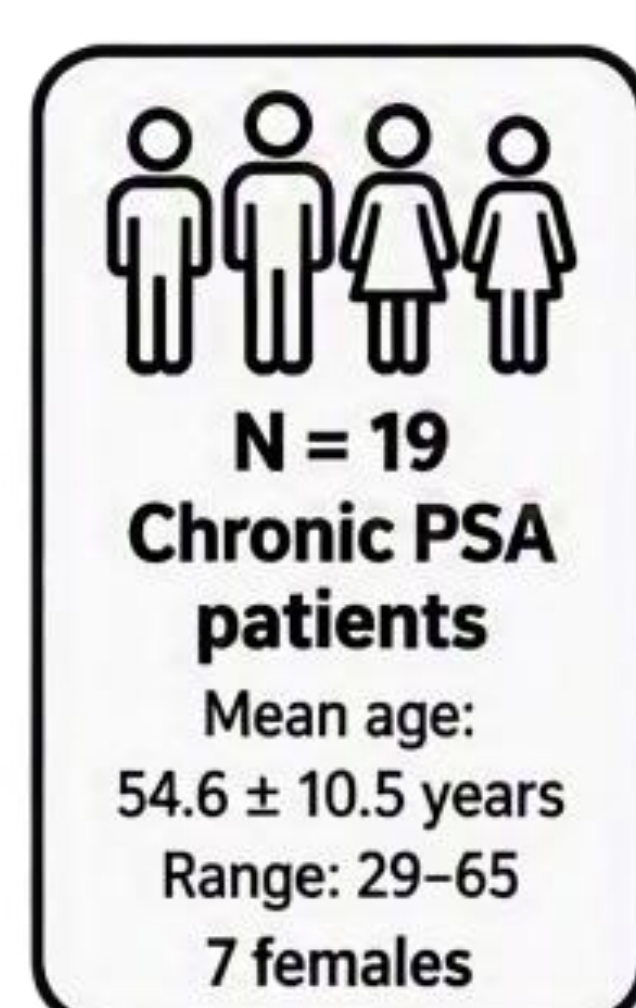


However, to our knowledge, no previous study has examined the effects of donepezil on word and pseudoword reading in chronic PSA.

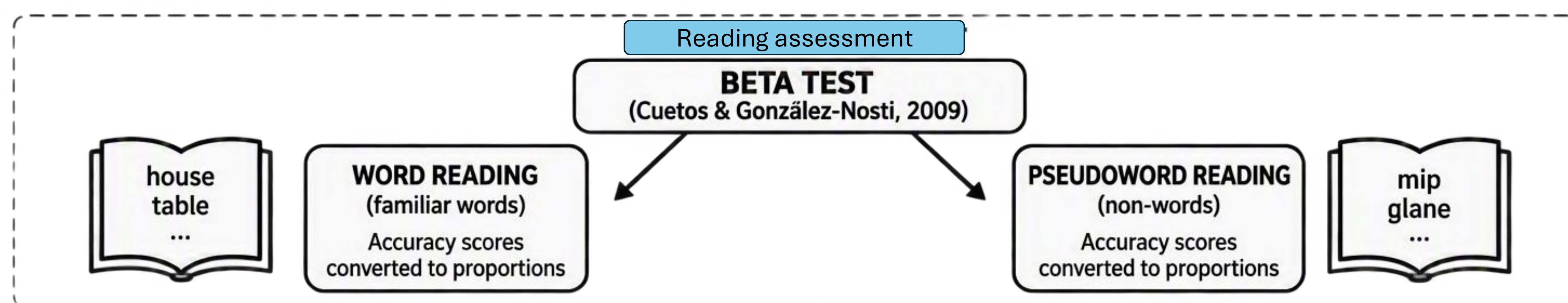
4. AIMS

- To investigate the effects of donepezil alone and in combination with intensive language therapy on WORD and PSEUDOWORD reading performance in chronic PSA.
- To determine whether baseline performance predicts treatment response.

METHODS



* REGIA: Rehabilitación Grupal Intensiva de la Afasia is the Spanish adaptation of the Intensive Language Action Therapy (ILAT, Difrancesco et al. 2012)

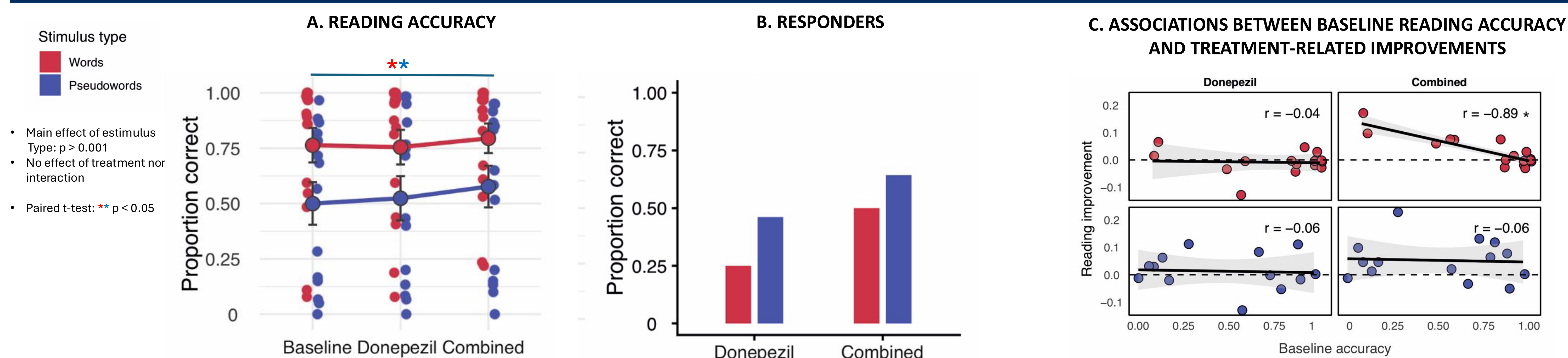


STATISTICAL ANALYSES:

Linear mixed-effects models examined the effects of time and stimulus type. Condition-specific changes were assessed using paired t-tests. Responder analyses were based on positive change scores ($\Delta > 0$).

Associations between baseline performance and treatment gains were evaluated using Pearson correlations, and multiple regression analyses were performed for word/pseudoword reading after the combined condition, including baseline performance and clinical variables as predictors.

RESULTS



CONCLUSIONS

- Combined language therapy improved reading in PSA, with treatment effects driven by individual differences rather than group-level changes, potentially reflecting variability in damage to reading and language brain networks (López-Barroso et al. 2020).
- Treatment gains were observed for both word and pseudoword reading (Fig. A), with improvements occurring after combined therapy (Fig. A and Fig. B).
- Word reading gains were strongly predicted by baseline performance, with greater improvement in more impaired patients (Fig. C), consistent with the lexical-semantic demands of the REGIA/ILAT.
- These findings highlight stimulus-specific recovery mechanisms and support individualized aphasia rehabilitation based on baseline performance.

REFERENCES



Funded by Projects:
PI16/01514, PID2021-
127617NA-I00,
RYC2020-029495-I

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